

Answer All Questions

1	a) Match the stress states in column A with the corresponding Mohr circles in column B		L3	
	A	B	CO1	4 marks
	1	1		
	2	2		
	3	3		
	4	4		
		5		
		6		
b) Establish the relation connecting shear force, bending moment, and intensity of force.			L2	
			CO3	2 marks
c) The bending moment diagram of a beam is shown. Can you draw the shear force diagram?			L3	
			CO2	3 marks
d) Draw the free body diagram of the beam shown in the figure at section x-x and find the shear force and bending moment at x-x.			L3	
			CO3	5 marks
Q2	The state of stress in MPa at a point in a stressed material is shown in the figure. Find the normal and shear stresses on a plane whose normal makes an angle of 20° with x-axis. Also find the principal stresses, orientation of principal planes, and maximum in-plane shear stress.		L4	
			CO4	12 marks
Q3.	Draw the shear force and bending moment diagrams for the beam loaded and supported as shown. What is the maximum bending moment developed in the beam? Where is it developed? Locate the point of contraflexure, if any.		L4	
			CO 2	12 marks
Q4.	For the beam shown in Q3, determine the slope and deflection at the centre of the span AB, and at the free end D.		L4	
			CO4	12 marks